

Aegis Series-7 Hydraulic Control System

Operator Manual

Document AEG-OM-700 | Revision 4 | Applies to software revision 3.2 and later

1. Overview

The Aegis Series-7 Hydraulic Control System (HCS) provides closed-loop hydraulic pressure regulation for industrial press lines. The system is built around the Hydraulic Power Unit (HPU), which supplies pressurized fluid to the press circuit under the supervision of the HCS Controller (PLC-03).

Throughout Aegis documentation the Hydraulic Power Unit may also be referred to as the **Hydraulic Unit** or the **HP unit**, depending on which department produced the document. All three terms refer to the same physical assembly.

2. Key Components

Reference	Component	Function
HPU	Hydraulic Power Unit	Generates and regulates hydraulic pressure
IV-21	Isolation Valve 21	Isolates the HPU from the press circuit
PS-04A	Pressure Sensor 04A	Reports HPU discharge pressure to PLC-03
PLC-03	HCS Controller	Supervises startup, monitoring, and alarms

Note: earlier Aegis units shipped with pressure sensor **PS-04**. As of software revision 3.2, PS-04 is superseded by PS-04A; see Engineering Change Notice ECN-1042 for details.

4. Operating Procedures

4.3 Starting the Hydraulic Power Unit

- Before starting, verify:
- Hydraulic fluid level is within the normal band on the sight glass.
 - Isolation valve **IV-21** is OPEN.
 - The emergency stop circuit is RESET.

WARNING: Do not start the Hydraulic Power Unit with the maintenance access panel removed. Rotating components inside the HPU housing are exposed when the panel is off.

- Startup sequence:
- Turn the POWER selector to ON.
 - Press START.
 - Verify discharge pressure, as reported by PS-04A, reaches **200 bar**. This is the normal operating pressure for software revision 3.2 and later; see ECN-1042 for the pressure change history.

5. Normal Operating Ranges

Normal HPU discharge pressure is **200 bar** on software revision 3.2 and later. On revisions prior to 3.2, normal discharge pressure was 180 bar, measured by the original PS-04 sensor. Do not compare live readings against

the 180 bar figure on a unit running revision 3.2 or later — the setpoint changed along with the sensor replacement.

6. Alarms

See the Aegis Alarm Reference (AEG-AL-700) for the full alarm list, including alarm A17 (Hydraulic Pressure Low), its possible causes, and required operator actions.

7. Controller Reset

Do not reset the PLC-03 controller while hydraulic pressure is above 50 bar. Resetting the controller under pressure can cause an uncommanded valve transition. Bleed pressure below 50 bar first, using the manual bleed procedure in Section 8, before performing a controller reset.